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 Studierichting: Informatica
 Oefeningenreeks 3D nr. 7.4

Bepaal extrema van $f(x) = \sin x + \frac{\sin 2x}{2} + \frac{\sin 3x}{3}$

$$\bullet f'(x) = \cos x + \cos(2x) + \cos(3x) = \cos x + \cos(3x) + \cos(2x)$$

$$\text{Simpson : } \left(\cos a + \cos b = 2 \cos \left(\frac{a+b}{2} \right) \cdot \cos \left(\frac{a-b}{2} \right) \right)$$

$$\Leftrightarrow f'(x) = 2 \cos \left(\frac{4x}{2} \right) \cdot \cos \left(\frac{2x}{2} \right) + \cos(2x)$$

$$\bullet f'(x) = 0 \Leftrightarrow 2 \cos(2x) \cos x + \cos(2x) = 0$$

$$\Leftrightarrow \cos(2x) \cdot (2 \cos x + 1) = 0$$

$$\text{dus } \cos(2x) = 0$$

$$\Leftrightarrow 2x = \frac{\pi}{2} + k\pi \Leftrightarrow x = \frac{\pi}{4} + \frac{k\pi}{2}$$

$$\text{of } 2 \cos x + 1 = 0$$

$$\Leftrightarrow \cos x = -\frac{1}{2} \Leftrightarrow x = \frac{2\pi}{3} + k\pi \text{ of } x = \frac{4\pi}{3} + k\pi$$

x	$\frac{\pi}{4}$		$\frac{2\pi}{3}$		$\frac{3\pi}{4}$		$\frac{5\pi}{4}$		$\frac{4\pi}{3}$		$\frac{7\pi}{4}$		
$f'(x)$	+	0	-	0	+	0	-	0	+	0	-	0	+
$f(x)$	$\nearrow$	M	$\searrow$	m	$\nearrow$	M	$\searrow$	m	$\nearrow$	M	$\searrow$	m	$\nearrow$

References